Supplement: Pure Relative Structured Matrix Approximation with Square-Root Matvecs

A Partial-Trace Covariance

This section proves the concentration lemma used in the main paper. We state a slightly more general accuracy form.

Lemma 1 (Partial-trace Gaussian covariance). *Let $C_1, \dots, C_q \in \mathbb{R}^{n \times m}$ satisfy*

$$K = (\langle C_i, C_j \rangle_F)_{i,j} \preceq I, \qquad \sum_{i=1}^{q} C_i C_i^\top \preceq \tau I, \tag{1}$$

where $\tau \geq 1$. For $g \sim N(0, I_m)$, let $D(g) = [C_1 g, \dots, C_q g] \in \mathbb{R}^{n \times q}$. There is a universal constant C such that, for every $\eta \in (0,1)$ and

$$k \geq \frac{C\tau \log^2(2q)}{\eta^2}, \tag{2}$$

independent $g_1, \dots, g_k$ satisfy

$$\left\| \frac{1}{k} \sum_{\ell=1}^{k} D(g_\ell)^\top D(g_\ell) - K \right\|_{\mathrm{op}} \leq \eta \tag{3}$$

with probability at least 0.99.

We use the following standard rectangular form of noncommutative Khintchine (Buchholz, 2001). The Gaussian statement follows from the same trace-moment proof as the Rademacher statement. It can also be obtained by self-adjoint dilation.

Lemma 2 (Rectangular Gaussian Khintchine). *For fixed real matrices $A_1, \dots, A_m$ and an integer $p \geq 1$,*

$$\begin{aligned} &\left(\mathbb{E} \left\| \sum_{a=1}^{m} g_a A_a \right\|_{S_{2p}}^{2p} \right)^{1/(2p)} \\ &\leq \sqrt{2p-1} \max \left\{ \left\| \left(\sum_a A_a A_a^\top \right)^{1/2} \right\|_{S_{2p}}, \right. \\ &\qquad\qquad \left. \left\| \left(\sum_a A_a^\top A_a \right)^{1/2} \right\|_{S_{2p}} \right\}. \end{aligned} \tag{4}$$

Here $\| \cdot \|_{S_r}$ denotes the Schatten r-norm.

Proof of Lemma 1. For each input coordinate $a \in [m]$, define

$$A_a = [C_1 e_a, \ldots, C_q e_a] \in \mathbb{R}^{n\times q}.$$

Then $D(g) = \sum_a g_a A_a$, and direct calculation gives

$$\sum_a A_a A_a^\top = \sum_i C_i C_i^\top \preceq \tau I, \qquad \sum_a A_a^\top A_a = K \preceq I. \tag{5}$$

Both matrices in (5) have trace $\operatorname{tr} K \le q$. Consequently,

$$\operatorname{tr}\left(\sum_a A_a A_a^\top\right)^p \le \tau^{p-1} q, \tag{6}$$

$$\operatorname{tr} K^p \le q. \tag{7}$$

Apply Lemma 2, use $\|M\|_{\mathrm{op}} \le \|M\|_{S_{2p}}$, and take $p \ge \log(2q)$. Since $q^{1/(2p)}$ is bounded by a universal constant and $\tau \ge 1$, we obtain

$$\left(\mathbb{E}\|D(g)\|_{\mathrm{op}}^{2p}\right)^{1/(2p)} \le C\sqrt{p\tau}. \tag{8}$$

Set $Z = D(g)^\top D(g)$, so $\mathbb{E}Z = K$. We next compute its centered matrix variance. For a unit vector $\theta \in \mathbb{R}^q$, let $C_\theta = \sum_i \theta_i C_i$. Since

$$\theta^\top Z^2 \theta = \sum_{i=1}^q \langle C_i g, C_\theta g\rangle^2,$$

the real Gaussian fourth-moment identity gives

$$\begin{aligned}
\theta^\top \mathbb{E}Z^2\theta &= \sum_i \left\{\langle C_i, C_\theta\rangle_F^2 + \operatorname{tr}[(C_i^\top C_\theta)^2]\right\} \\
&\quad + \sum_i \|C_i^\top C_\theta\|_F^2 \\
&\le \theta^\top K^2\theta + 2\sum_i \|C_i^\top C_\theta\|_F^2 \\
&= \theta^\top K^2\theta + 2\operatorname{tr}\left(C_\theta^\top \sum_i C_i C_i^\top C_\theta\right) \\
&\le \theta^\top (K^2 + 2\tau K)\theta.
\end{aligned} \tag{9}$$

The first equality uses $\mathbb{E}(g^\top M g)^2 = (\operatorname{tr} M)^2 + \operatorname{tr}(M^2) + \operatorname{tr}(MM^\top)$. Because $\mathbb{E}(Z-K)^2 = \mathbb{E}Z^2 - K^2$, equation (9) implies

$$\|\mathbb{E}(Z-K)^2\|_{\mathrm{op}} \le 2\tau. \tag{10}$$

Let $X_\ell = Z_\ell - K$ for independent copies Z_ℓ. The variance parameter of $\sum_\ell X_\ell$ is at most $2k\tau$. We also need the large-deviation parameter

$$L = \left(\mathbb{E}\max_{\ell\le k}\|X_\ell\|_{\mathrm{op}}^2\right)^{1/2}.$$

For any integer $p \ge \log(2q)$,

$$\begin{aligned}
\|\|X_\ell\|_{\mathrm{op}}\|_{L_{2p}} &\le \|\|D(g_\ell)\|_{\mathrm{op}}^2\|_{L_{2p}} + 1 \\
&= \|\|D(g_\ell)\|_{\mathrm{op}}\|_{L_{4p}}^2 + 1 \le Cp\tau.
\end{aligned}$$

The elementary maximum bound gives

$$\begin{aligned}
L &\le \left(\mathbb{E}\max_{\ell\le k}\|X_\ell\|_{\mathrm{op}}^{2p}\right)^{1/(2p)} \\
&\le k^{1/(2p)}\max_{\ell\le k}\|\|X_\ell\|_{\mathrm{op}}\|_{L_{2p}}.
\end{aligned}$$

Taking $p = \lceil \log(2qk) \rceil$ yields

$$L \le C\tau \log(2qk). \tag{11}$$

The expected-norm theorem for independent centered random matrices (Tropp, 2015, Theorem I), applied in dimension q, now gives

$$\left(\mathbb{E} \left\| \frac{1}{k} \sum_{\ell=1}^{k} X_\ell \right\|_{\mathrm{op}}^2 \right)^{1/2} \le C\sqrt{\frac{\tau \log(2q)}{k}} + \frac{C\tau \log(2q) \log(2qk)}{k}. \tag{12}$$

Let $k_0 = C_0 \tau \log^2(2q)/\eta^2$. Since τ may be replaced by $\min\{\tau, q\}$, we may assume $\tau \le q$. At $k = k_0$,

$$\log(2qk) = O(\log(2q) + \log(1/\eta)).$$

The first term of (12) is at most $C\eta/\sqrt{\log(2q)}$. For the second term, use $\eta \log(1/\eta) \le 1/e$. Increasing C_0 makes the root second moment at most $\eta/10$. For $k \ge k_0$, the first term decreases and $\log(2qk)/k$ also decreases. Markov's inequality proves (3) with probability at least 0.99. □

B Upper Bound and Amplification

We first record basis invariance and observability.

Lemma 3 (Basis invariance)**.** *The partial traces $S_L = \sum_i B_i B_i^\top$ and $S_R = \sum_i B_i^\top B_i$ do not depend on the choice of Frobenius-orthonormal basis of $\mathcal{L}$.*

Proof. Any two orthonormal bases are related by an orthogonal matrix U. If $B_i' = \sum_j U_{ij} B_j$, then

$$\sum_i B_i' B_i'^\top = \sum_{j,k} (U^\top U)_{jk} B_j B_k^\top = \sum_j B_j B_j^\top.$$

The right partial trace is identical after transposition. □

Lemma 4 (Observable objective)**.** *Let $P = VV^\top$, where $V \in \mathbb{R}^{n\times s}$ has orthonormal columns. The responses $A^\top V$ and AG determine*

$$\|P(A - B)\|_F^2 + k^{-1}\|Q(A - B)G\|_F^2$$

for every known B.

Proof. The left responses determine $PA = V(A^\top V)^\top$. Therefore PAG and $QAG = AG - PAG$ are known. All terms involving B are known from the basis. □

Lemma 5 (Offset variance)**.** *Suppose $\sum_i C_i C_i^\top \preceq \lambda I$. For fixed M, define*

$$\xi_i = \frac{1}{k} \sum_{\ell=1}^{k} \{\langle M g_\ell, C_i g_\ell \rangle - \langle M, C_i \rangle_F\}.$$

Then $\mathbb{E}\|\xi\|_2^2 \le 2\lambda \|M\|_F^2/k$.

Proof. For $N = M^\top C_i$, the Gaussian identity $\mathrm{Var}(g^\top N g) = \mathrm{tr}(N^2) + \mathrm{tr}(NN^\top)$ gives

$$\mathrm{Var}(\langle Mg, C_i g \rangle) \le 2\|M^\top C_i\|_F^2.$$

Summing over i and using independence across ℓ gives

$$\mathbb{E}\|\xi\|_2^2 \le \frac{2}{k}\sum_i \|M^\top C_i\|_F^2$$
$$= \frac{2}{k}\operatorname{tr}\left(M^\top \sum_i C_i C_i^\top M\right) \le \frac{2\lambda}{k}\|M\|_F^2.$$

□

Theorem 6 (One-orientation upper bound). *Fix s and let $\lambda = \lambda_{s+1}^L > 0$. For every $\gamma \in (0,1)$, if*

$$k \ge C\left[\max\{1,\lambda\}\log^2(2q) + \frac{\lambda}{\gamma}\right], \tag{13}$$

then the estimator in the main paper satisfies

$$\|\widehat{B} - B_\star\|_F^2 \le \gamma \mathrm{OPT}^2 \tag{14}$$

with probability at least 19/20.

Proof. Let $C_i = QB_i$ and $D(g) = [C_1 g, \ldots, C_q g]$. Its Gram matrix $K = (\langle C_i, C_j\rangle_F)$ satisfies $K \preceq I$, while

$$\sum_i C_i C_i^\top = QS_L Q \preceq \lambda I.$$

Set $\tau = \max\{1,\lambda\}$. The Hessian of the objective in the orthonormal coordinates of $\mathcal{L}$ is

$$H = H_P + \frac{1}{k}\sum_{\ell=1}^k D(g_\ell)^\top D(g_\ell),$$
$$(H_P)_{ij} = \langle PB_i, PB_j\rangle_F.$$

Because $H_P + K = I$, Lemma 1 with $\eta = 1/2$ implies

$$H \succeq \frac{1}{2} I \tag{15}$$

except with probability 1/100.

Write $R = A - B_\star$. At $B_\star$, the ith normal-equation offset is

$$h_i = \langle PR, PB_i\rangle_F + \frac{1}{k}\sum_{\ell=1}^k \langle QRg_\ell, QB_i g_\ell\rangle.$$

Orthogonality gives

$$\langle PR, PB_i\rangle_F + \langle QR, QB_i\rangle_F = \langle R, B_i\rangle_F = 0.$$

Thus h is the centered vector in Lemma 5 with $M = QR$. In particular,

$$\mathbb{E}\|h\|_2^2 \le \frac{2\lambda}{k}\mathrm{OPT}^2. \tag{16}$$

By Markov's inequality and a sufficiently large constant in (13),

$$\|h\|_2^2 \le \frac{\gamma}{4}\mathrm{OPT}^2 \tag{17}$$

except with probability 1/25.

Let d be the coefficient vector of $\widehat{B} - B_\star$. On (15), the objective is strictly convex and its normal equation is $Hd = h$. Therefore

$$\|\widehat{B} - B_\star\|_F^2 = \|d\|_2^2 \le 4\|h\|_2^2 \le \gamma\mathrm{OPT}^2.$$

A union bound gives success probability at least $1 - 1/100 - 1/25 = 19/20$. □

If $\lambda = 0$, then $QB_i = 0$ for all i. The exact part of the objective has Hessian I and recovers $B_\star$ without random queries. The profile in the main paper follows by taking $\gamma = 2\epsilon$ in Theorem 6 and applying

$$\|A - \widehat{B}\|_F^2 = \mathrm{OPT}^2 + \|\widehat{B} - B_\star\|_F^2.$$

We next prove amplification without estimating OPT or any residual norm.

Lemma 7 (Metric median amplification). *Let $\widehat{B}_1, \ldots, \widehat{B}_r$ be independent candidates, where r is odd and each candidate satisfies*

$$\|\widehat{B}_i - B_\star\|_F^2 \le \gamma \mathrm{OPT}^2 \tag{18}$$

with probability at least 19/20. Let $h = (r+1)/2$. For each i, let ρ_i be the hth smallest value among $\{\|\widehat{B}_i - \widehat{B}_j\|_F : j \in [r]\}$. Return a candidate minimizing ρ_i. There is a universal C such that $r \ge C \log(1/\delta)$ implies

$$\|\widehat{B} - B_\star\|_F^2 \le 9\gamma \mathrm{OPT}^2 \tag{19}$$

with probability at least $1 - \delta$.

Proof. A Chernoff bound shows that at least h candidates satisfy (18) with probability at least $1-\delta$. Condition on this event. Every good candidate has at least h good candidates within distance $2\sqrt{\gamma}\mathrm{OPT}$, so its radius is at most this value. The selected candidate has radius no larger. Its radius ball contains at least h candidates, and the good set also has at least h members. Since $2h > r$, the two sets intersect. If $\widehat{B}_j$ is in their intersection, then

$$\|\widehat{B} - B_\star\|_F \le \|\widehat{B} - \widehat{B}_j\|_F + \|\widehat{B}_j - B_\star\|_F \le 3\sqrt{\gamma}\mathrm{OPT}.$$

□

Take $\gamma = 2\epsilon/9$ in Lemma 7. Then

$$\|A - \widehat{B}\|_F^2 \le (1 + 2\epsilon)\mathrm{OPT}^2 \le (1+\epsilon)^2 \mathrm{OPT}^2.$$

The same exact responses $A^\top V$ are reused for all candidates. Only the Gaussian stage is repeated, proving the confidence statement in the main paper.

C Lower Bounds

C.1 Exact-recovery dimension bound

Proposition 8. *For $q = r^2$, some q-dimensional linear family requires at least r adaptive two-sided queries for any finite multiplicative approximation guarantee with positive worst-case success probability.*

Proof. Take $\mathcal{L} = \mathbb{R}^{r\times r}$ and draw A from a nondegenerate Gaussian distribution on this space. Fix a deterministic adaptive algorithm using $t < r$ queries. Conditional on any transcript up to a given round, the next query vector is fixed. Its response gives at most r scalar linear constraints on the r^2 entries of A, whether the orientation is A or $A^\top$. Sequential Gaussian conditioning shows that after t rounds the conditional law remains Gaussian on an affine space of dimension at least $r^2 - tr > 0$. It is non-atomic. No transcript-measurable output equals A with positive probability.

For every $A \in \mathcal{L}$, the optimum residual is zero. Any finite multiplicative guarantee must output A exactly. Therefore every deterministic algorithm has zero average success under the Gaussian prior. Yao's principle rules out a randomized algorithm with positive worst-case success. □

C.2 One-parameter accuracy bound

We give the deferred-decisions argument and constants for the one-parameter proposition in the main paper. Write $G \sim \mathrm{GOE}_d$ when the diagonal entries of G are independent $N(0, 2)$ variables and its entries strictly above the diagonal are independent $N(0, 1)$ variables, with symmetry below the diagonal.

Lemma 9 (Adaptive GOE residual)**.** *Fix a deterministic adaptive algorithm that queries a symmetric matrix by matvecs. After t linearly independent queries to $G \sim \mathrm{GOE}_d$, there is a transcript-measurable orthogonal matrix U_t such that, conditionally on the transcript,*

$$U_t^\top G U_t = \Delta_t + \begin{bmatrix} 0_{t\times t} & 0 \\ 0 & H_t \end{bmatrix}, \qquad H_t \sim \mathrm{GOE}_{d-t}, \tag{20}$$

where Δ_t is transcript measurable and H_t is independent of the transcript.

Proof. Repeated queries and components in the span of previous queries reveal no new information, so Gram–Schmidt reduces to orthonormal query directions. We prove the claim by induction. It is immediate for $t = 0$. Suppose it holds after t queries. In the coordinates of (20), the next new query has a nonzero component in the last $d - t$ coordinates. Conditional on the transcript, rotate only those coordinates so that its normalized new component is the first coordinate. Rotational invariance leaves H_t GOE. The response reveals the first row and column of this rotated H_t. By the independent-entry definition of GOE, its remaining principal block is an independent GOE_{d-t-1} matrix. Absorb every revealed entry into Δ_{t+1}. This proves the induction, including when each query is chosen from all earlier responses. ☐

Proposition 10 (One-parameter accuracy lower bound)**.** *For all sufficiently small $\epsilon > 0$, some one-dimensional linear family requires $\Omega(1/\sqrt{\epsilon})$ adaptive two-sided queries for success probability at least $2/3$ at relative error $1 + \epsilon$.*

Proof. Set $c = 1/100$, let $d = \lfloor c/\sqrt{\epsilon} \rfloor$, and take

$$\mathcal{L} = \mathrm{span}\{I_d/\sqrt{d}\}, \qquad A \sim \mathrm{GOE}_d\,.$$

The input is symmetric, so a transpose query reveals the same response as a forward query. It is enough by Yao's principle to fix a deterministic adaptive algorithm. Its output is $\widehat{\beta} I_d/\sqrt{d}$, while the optimal coefficient is $\beta_\star = \mathrm{tr}(A)/\sqrt{d}$.

After $t \le d/2$ informative queries, Lemma 9 shows that the unknown part of $\beta_\star$, conditionally on the transcript, is

$$\frac{\mathrm{tr}(H_t)}{\sqrt{d}} \sim N\left(0, \frac{2(d-t)}{d}\right).$$

Its conditional variance is at least one. The density of a Gaussian with variance at least one is everywhere at most $1/\sqrt{2\pi}$. Hence, for every transcript-measurable estimate $\widehat{\beta}$ and every $a > 0$,

$$\Pr\bigl\{|\widehat{\beta} - \beta_\star| \le a \mid \text{transcript}\bigr\} \le \frac{2a}{\sqrt{2\pi}}. \tag{21}$$

We have $\mathbb{E}\|A\|_F^2 = d(d+1)$, so Markov's inequality gives the event

$$\mathrm{OPT}^2 \le \|A\|_F^2 \le 10d(d+1) \tag{22}$$

with probability at least 0.9. For $\epsilon \le 1$, a successful $(1+\epsilon)$ approximation must obey

$$\begin{aligned}(\widehat{\beta} - \beta_\star)^2 &\le \bigl((1+\epsilon)^2 - 1\bigr)\mathrm{OPT}^2 \\ &\le 3\epsilon \mathrm{OPT}^2.\end{aligned}$$

On (22), the right side is at most $30\epsilon d(d+1) \le 60c^2$. Apply (21) with $a = \sqrt{60}c$. The total success probability is at most

$$\frac{1}{10} + \frac{2\sqrt{60}c}{\sqrt{2\pi}} < \frac{1}{6}.$$

For sufficiently small ϵ, we also have $d \ge c/(2\sqrt{\epsilon})$. Therefore success probability $2/3$ requires more than $d/2 = \Omega(1/\sqrt{\epsilon})$ queries under this input distribution. Yao's principle gives the claimed worst-case lower bound for randomized algorithms. ☐

C.3 Joint accuracy bound

We state the source theorem in the form needed here. It is Theorem 2 of Amsel et al. (2026), together with the dimension choice in its proof.

Proposition 11 (Fixed-sparsity Wishart lower bound)**.** *Fix $\alpha \in (0,1)$. There are constants $a, b, c_0, c_1 > 0$ such that, for every integer $u \geq 1$ and $\epsilon < c_0$, one can choose*

$$a\frac{u}{\epsilon} \leq d \leq b\frac{u}{\epsilon} + 1 \tag{23}$$

and a symmetric Wishart distribution on $d \times d$ matrices with this property. For every mask having between αu and u entries in every row and column, any possibly randomized adaptive algorithm using fewer than $c_1 u/\epsilon$ matvec queries has probability at most $1/25$ of returning a $(1+\epsilon)$ fixed-mask approximation.

The source theorem is stated for forward queries. Its hard matrix is symmetric, so allowing transpose queries does not enlarge the transcript.

Theorem 12 (Joint lower bound for linear families)**.** *There are universal constants $c, c_0, c_2 > 0$ such that, whenever $\epsilon < c_0$ and $q\epsilon \geq c_2$, some q-dimensional linear family requires at least $c\sqrt{q/\epsilon}$ possibly adaptive two-sided queries for constant-probability $(1+\epsilon)$ approximation.*

Proof. For each u and d in Proposition 11, choose a cyclic u-regular mask $S \in \{0,1\}^{d\times d}$. This exists after reducing c_0 if necessary so that $d \geq u$. The coordinate family

$$\mathcal{L}_0 = \{B : B = S \circ B\}$$

has dimension

$$q_0 = du = \Theta(u^2/\epsilon). \tag{24}$$

Proposition 11 gives query lower bound

$$\Omega(u/\epsilon) = \Omega(\sqrt{q_0/\epsilon}). \tag{25}$$

Given q with $q\epsilon$ above a large enough constant, choose $u = \lfloor c_3\sqrt{q\epsilon}\rfloor$ for a sufficiently small universal c_3. Equations (23) and (24) ensure $c_4 q \leq q_0 \leq q$ for a universal $c_4 > 0$. Rounding changes only constants.

Let $p = q - q_0$. On a disjoint $p \times p$ block, let $\mathcal{D}_p$ be the diagonal coordinate family, and define

$$\mathcal{L} = \{B_0 \oplus D : B_0 \in \mathcal{L}_0,\ D \in \mathcal{D}_p\}.$$

This family has dimension exactly q. Embed a hard input as $A' = A_0 \oplus 0$. Its best approximation in $\mathcal{L}$ is $(S \circ A_0) \oplus 0$, and its optimum residual equals the original one.

Suppose an algorithm succeeds for $\mathcal{L}$. A query to A' with vector (x, z) returns $(A_0 x, 0)$, and a transpose query is identical. Thus every query can be simulated using one query to A_0. The algorithm's output has the form $\widehat{B}_0 \oplus \widehat{D}$. Success implies

$$\begin{aligned}\|A_0 - \widehat{B}_0\|_F^2 &\leq \|A_0 - \widehat{B}_0\|_F^2 + \|\widehat{D}\|_F^2 \\ &\leq (1+\epsilon)^2\|A_0 - S \circ A_0\|_F^2.\end{aligned}$$

Hence restriction to the first block solves the source problem. By (25) and $q_0 \geq c_4 q$, the required number of queries is $\Omega(\sqrt{q/\epsilon})$. □

D Profiles for Basic Families

Coordinate subspaces. For a binary mask S, use basis matrices E_{ij} at its nonzero locations. Then

$$\begin{aligned}S_L &= \mathrm{diag}(d_1^{\mathrm{row}}, \ldots, d_n^{\mathrm{row}}), \\ S_R &= \mathrm{diag}(d_1^{\mathrm{col}}, \ldots, d_m^{\mathrm{col}}).\end{aligned}$$

The spectral split exactly queries selected high-degree rows or columns and uses a Gaussian fit on the remainder.

Full rectangular block. For all matrices supported on an $a \times b$ block, $S_L = bI_a$ on the row block and $S_R = aI_b$ on the column block. Taking $s = a$ in the left profile or $s = b$ in the right profile recovers the block exactly in $\min\{a, b\}$ queries. For $a = b = \sqrt{q}$, this matches Proposition 8.

Row star. For a single active row with q independent entries, $S_L = qee^\top$. One left query in direction e observes the entire structured component and gives exact recovery.

Diagonal family. For the $q \times q$ diagonal family, $S_L = S_R = I_q$. Taking $s = 0$ gives $O(\log^2(2q) + 1/\epsilon)$ queries. The specialized rowwise inverse-Wishart analysis removes the logarithmic term (Amsel et al., 2026). This example shows that the universal profile is adaptive to diffuse leverage, even though the general matrix-concentration proof is not logarithmically sharp.

References

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